



SERIES

FOURTH PRIMARY, FIRST TERM 2027

SCIENCE

By A Group of Supervisors

4



EL-MOASSER
Interactive E-learning
Application



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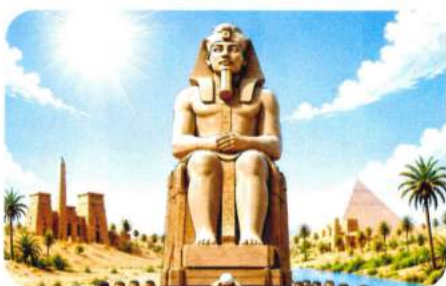
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CURRICULUM OBJECTIVES

Chapter 1 By the End of this chapter, the pupil should be able to:

- 1
 1. Distinguish between living organisms and nonliving things.
 2. Explore where living organisms are found.
- 2
 1. Use a magnifying glass to observe details of some living organisms.
 2. Correctly prepares a record card for some living organisms.
- 3
 1. Recognize the concept of classification.
 2. Classify living organisms by color, shape, and size.
- 4
 1. Compare the characteristics of the Nile River bank environment and the desert environment and identifies the living organisms found in each.
 2. Recognize the concept of adaptation.
 3. Identify physical features that help desert plants, such as acacia, survive in their environment.
- 5
 1. Recognize physical features that help desert animals live and survive in their environment.
 2. Recognize clever ways of living that help desert animals live and survive in their environment.

Chapter 2 By the End of this chapter, the pupil should be able to:

- 1
 1. Describe the stages of insect growth and the order of its life cycle.
 2. Identify the egg and larva stages.
 3. Explain the importance of molting during larva growth.
- 2
 1. Identify the pupa and adult stages.
 2. Describe the changes that occur as an insect becomes an adult.
 3. Relate the stages of insect growth to the continuation of its life cycle.
- 3
 1. Identify the body structure of an adult insect.
 2. Compare insects and spiders in body structure and number of legs.

Chapter 3 By the End of this chapter, the pupil should be able to:

- 1
 1. Explore seed diversity and differences in shape, color, and size according to plant type.
 2. Identify the best time to sow the seeds of some plants, such as sunflower and fava bean.
 3. Apply the correct practical steps for sowing seeds in soil.
- 2
 1. Identify the stages of plant growth after sprouting.
 2. Distinguish between seed leaves [cotyledons] and true leaves in shape, color, and number.
- 3
 1. Identify the stages of flower and fruit formation in plants.
 2. Explain the role of flowers in forming fruits and seeds.
 3. Relate fruit formation to the continuation of the plant life cycle.
- 4
 1. Identify the three main parts of a plant and the importance of each.
 2. Examine the shape and color of plant roots after washing away the soil.



Chapter 4 By the End of this chapter, the pupil should be able to:

1. Identify light, some of its sources, and how it travels.
2. Identify the concept of light reflection.
3. Distinguish between light reflection from smooth and rough surfaces.
2. 1. Identify how to gather sunlight using a magnifying glass.
2. Identify the properties of a sunlight focal point made with a magnifying glass.
3. 1. Identify how shadows form.
2. Determine the relationship between the Sun's position and the direction in which a shadow extends.
4. 1. Describe the Sun's movement during the day.
2. Explain why a shadow's length and direction change during the day.
3. Identify how a sundial is used to estimate time.
5. 1. Infer the relationship between the amount of sunlight, ground-surface temperature, and moisture.
2. Compare sunny and shaded places.
3. Identify how to measure temperature using a regular thermometer or an infrared thermometer.

Chapter 5 By the End of this chapter, the pupil should be able to:

1. 1. Identify how sound is produced.
2. Identify the relationship between an object's vibration and the sound it produces.
3. Experimentally verifies the relationship between an object's vibration and the production of sound.
4. Explain how sound travels through the surrounding medium.
2. - Infer the relationship between the amount of vibration and sound loudness.
3. 1. Conclude that sound travels through solids and air.
2. Infer how different materials affect the clarity of sound.

Chapter 6 By the End of this chapter, the pupil should be able to:

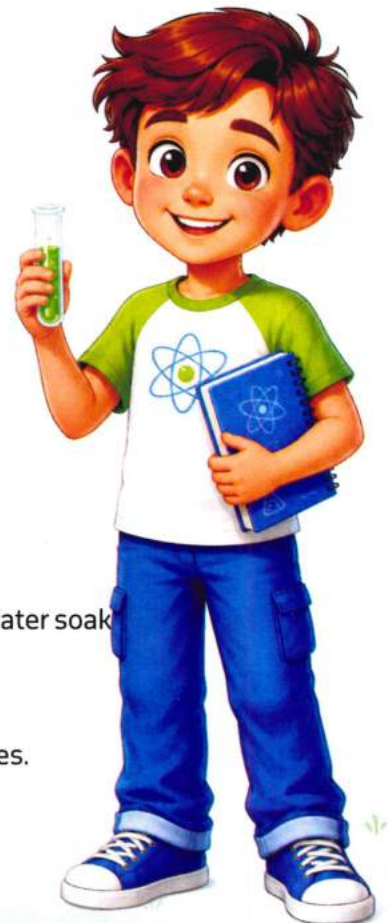
1. 1. Recognize the concept of force.
2. Compare how strong and weak winds affect the motion of objects.
2. - Infer the relationship between wind strength and the surface area exposed to the wind.

Chapter 7 By the End of this chapter, the pupil should be able to:

1. - Identify how the amount a rubber band is stretched affects the motion of objects.
2. - Infer how increasing the number of rubber bands affects the motion of objects.

Chapter 8 By the End of this chapter, the pupil should be able to:

1. 1. Infer the relationship between soil particle size and the rate at which water soaks through it.
2. Compare sandy soil and clay soil.
2. - Recognize how ancient Egyptians were able to move giant stone statues.



CHAPTER ONE

Nature in Our Surroundings

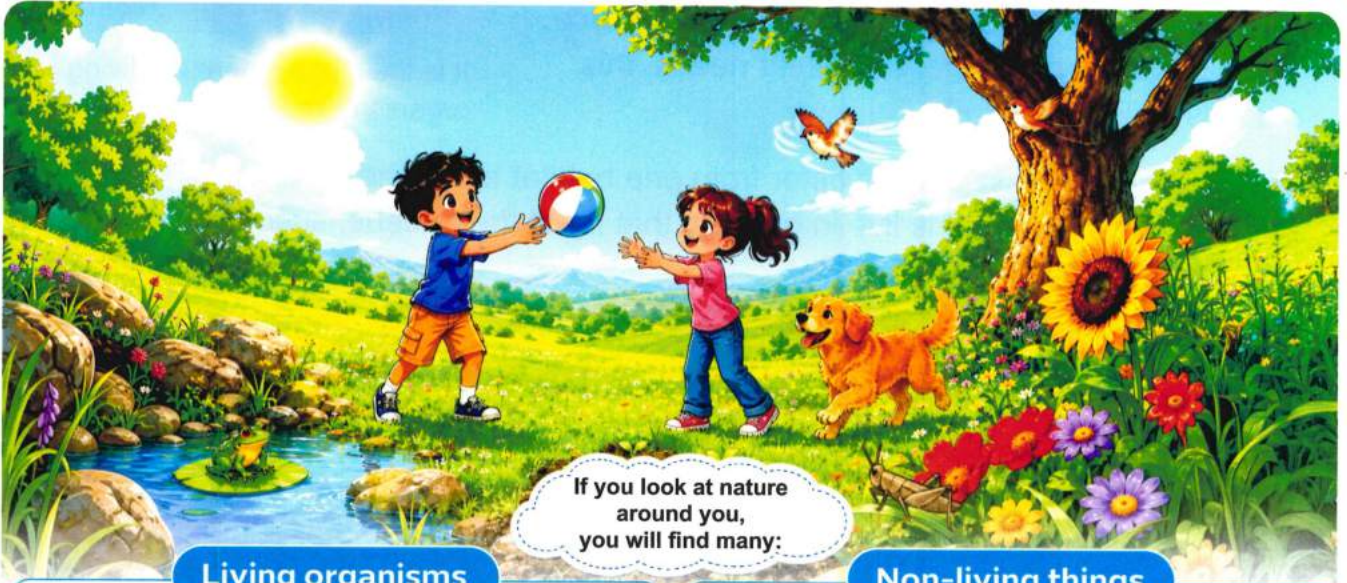


CHAPTER

1

- Lesson 1 Let's Find Living Organisms
- Lesson 2 How to Observe
- Lesson 3 Colors and Shapes of Living Organisms
- Lesson 4 Different Places, Different Living Organisms
- Lesson 5 Desert Living Organisms

Let's Find Living Organisms



Living organisms

Such as



Humans

Animals

Plants

Non-living things

Such as



Air

Water

Rocks

Characteristics of Living Organisms

- Living organisms differ from non-living things because they have several characteristics, including:



Feeding



Growth



Breathing



What is meant by Living organisms?

Everything that can feed, grow, and breathe, such as humans, animals, and plants.

nature
characteristics
living organisms

الطبيعة
خصائص
الكائنات الحية

non-living things
breathing
feeding

الأشياء غير الحية
التنفس
التغذية

human
growth

الإنسان
النمو

Habitats of Living Organisms

► Many living organisms live in places near us, such as the schoolyard.

► All living organisms cannot live in the same place. They look for a place that suits their needs, this place is called "**Habitat**".

What is meant by Habitat?

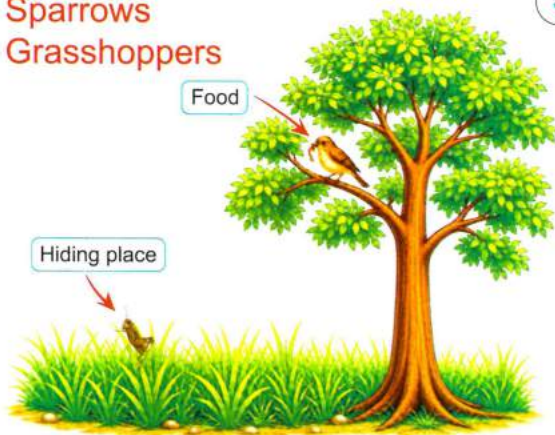
It is the place where the living organism lives.

► The kinds of living organisms differ from one habitat to another **G.R?**
Because living organisms live in habitats that meet their needs, such as water, food, and shelter.

Living organisms may be found :

1 Near grass or trees **G.R?**

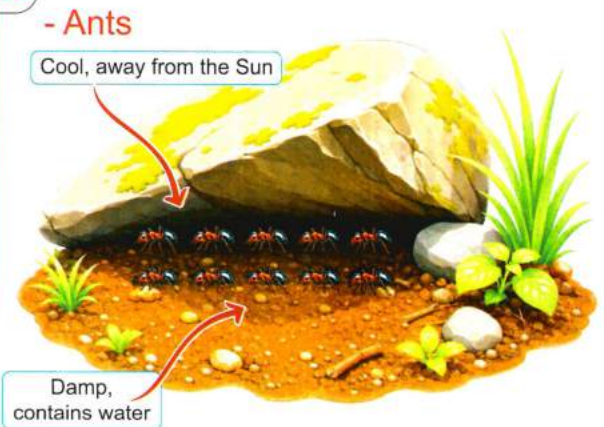
- Because they are:
- Places with food.
 - Places easy to hide in.
 - Sparrows
 - Grasshoppers



Such as

2 Under stones, in the shade of rocks, or under the ground **G.R?**

- Because they are:
- Damp places that contain water.
 - Cool places away from the Sun.



Living Organisms and the environment

► Living organisms live in ways that suit the environment that they live in. For example:
Some living organisms can live in hot, dry places such as deserts by living:

- In the shade of rocks. **or** • Under the ground. **G.R?**
To avoid the intense heat of the Sun in desert during the day.



! For illustration only

Environment is everything around a living organism, including the other living organisms and non-living things.

avoid
suits
habitat

يتجنب shelter
يناسب hide
موطن sparrows

مأوى grasshoppers
يختفي damp
العصافير ants

الجراد
رطب
نمل

Activity

Exploring the Habitats of Some Living Organisms

► To discover where living organisms in the following figures live.

We should follow these steps:



1 Guess



- Predict places where living organisms might be found, **such as:**
- Flower beds.
 - Roots of big trees.
 - Gaps between rocks.

2 Search



- Go outside and look for living organisms in the places you predicted.
- If you find one, watch it quietly and carefully.

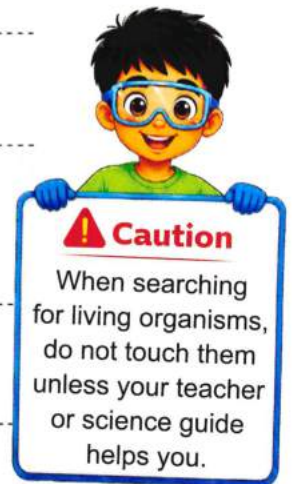
3 Record



- When you find a living organism, record both:
- The name of the living organism.
 - The kind of place where you found it.

► By following the previous steps, you can explore the habitats of these living organisms, as shown in the following table:

 Living organism	Place (habitat) 
Sparrow 	- On trees.
Spider 	- Flower beds. - Gaps between rocks.
Lizard 	- Under the shade of rocks. - Under stones. - On hot rocks.
Scorpion 	- Under stones. - Gaps between rocks.
Dung beetle 	- Deserts. - Farmland near the Nile River.



exploring
guess
record

استكشاف predict
تخمين flower beds
تسجيل roots

يتنبأ gaps
أحواض الزهور shade
حذور farmland

فجوات
ظل
الأراضي الزراعية

Questions on Chapter 1

LESSON

1

Let's Find Living Organisms

Questions signed by  are taken from the "Performances and Assessments Book"

First Questions of El-Moasser Book

1 Complete the following sentences:

- ① Humans and animals are examples of
- ②  Both and are non-living things.
- ③  Living organisms differ from non-living things in many characteristics such as , and
- ④ The place where a living organism lives is called a
- ⑤ Some living organisms can live under or the shade of

2 Put [✓] or [X]:

- ① Birds and ants are considered from animals. ()
- ② Living organisms can be found in more than one habitat around us. ()
- ③ Grasshoppers live near grass or trees. ()
- ④ All living organisms can live in hot, dry places such as deserts. ()

3 Give reasons for the following:

- ① Trees and frogs are living organisms.
- ② Rabbits prefer live in places near grass and trees.
- ③ Some living organisms prefer to live under stones.
- ④ The kinds of living organisms you find differ depending on the location that you finding in.

4 What happens when a living organism is placed in an environment that does not meet its needs?

.....

5 Mention:

- ① The steps for exploring the habitats of living organisms, in order.
.....
- ② Examples of places where living organisms may be found.
.....

6 Based on what you have learned, **where** can you find the following living organisms?



①



②



③



④

Second Questions of the School Book

A Try:

1 Choose the correct words from below to fill in the blanks:

(damp – rocks – grass)

Where do these living organisms often live?

- ① Grasshoppers are often in the
- ② Ants are often in places under stones.
- ③ Lizards are sometimes seen on hot

2 Complete: Living organisms live in that fit them. So, the kinds of living organisms you find are different depending on the location.

B Exercise:

1 Choose all the correct answers:

Where is it easy to find living organisms in a hot, dry place like Egypt?

- ① Right in the middle of the hot sand in the Sun.
- ② Cool places in the shadow of rocks or big stones.
- ③ At the roots where a little bit of grass grows.

2 Choose the correct words from below to fill in the blanks:

(dampness (water) – hide – food)

Where living organisms are?

- ① Near grass or trees: Places with or places easy to
- ② Under stones or shadows of rocks: Places with or cool places away from the Sun.

How to Observe

Rules for observation

- ▶ When looking closely to living organisms, you must observe its:

Shape and **Color** and **Size**

- ▶ To see details of some living organisms, we use a **Magnifying glass** **Then** write a **Record card**.



First How to Use a Magnifying Glass

- ▶ You can use a magnifying glass in one of two ways:

What is meant by Magnifying glass?

It is a tool used to help you see details of some living organisms. (Such as insects and plants).

1 When the object can be moved
(Such as a small leaf)

Hold the magnifying glass close to your eye, **then**:

Move **the object** back and front to get a clear image.



2 When the object cannot be moved
(Such as a tree trunk)

Move **your face with the magnifying glass** back and front to get a clear image.



Caution

- ▶ Never look directly at the Sun with a magnifying glass **G.R. ?** Because it can harm your eyes.
- ▶ Wear gloves while observing the living organism **G.R. ?** To protect your hands from getting polluted.



observation

size

magnifying glass

الملاحظة

حجم

عدسة مكبرة

record card

tree trunk

insects

بطاقة الملاحظة

جذع شجرة

حشرات

gloves

polluted

leaf

قفازات

يتلوث

ورقة نبات

► You can prepare a record card for a living organism as the following:

- 1 Write the topic.
- 2 Write date, grade and your name.
- 3 Record what you found using:
 - (1) A sketch with thin, clear lines and no shading.
 - (2) Sentences to describe
 - The characteristics of the living organism you observed, such as its size, color and shape.
 - The place (location) where you found it.
- 4 Write what you learned or write the questions you are thinking about.

Activity

Observing a Living Organism Closely

Steps

1 Use a magnifying glass to observe both:

Dung beetle

► Look at the insect's shape and body parts, **such as:**

- Small hairs.
- Leg joints.

Plant leaf

► Look at the leaf and observe the lines (veins) on it.

prepare
topic
sketch

يُحضّر / يَعدّ describe
عنوان location
رسم توضيحي joints

يصف veins
مكان / موقع dung beetle
مفاصل

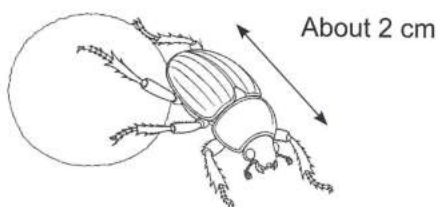
العروق
خنافس الروث

2 Prepare a record card for each one.

Observation

Dung beetle

Name: Grade: Date:

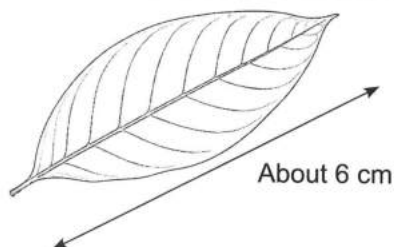


Size	About 2 cm
Color	Black (Blackish).
Shape	Oval-shaped with long back legs.
Location	On the ground near the Nile River.

It rolled a perfectly round ball of dung backward while standing on its head.

Plant leaf

Name: Grade: Date:



Size	About 6 cm
Color	Dark green.
Shape	Oval.
Location	Tree branches.

A prominent long vein is found at the center of the leaf, with a network of smaller veins branching from it on both sides.

For more information about

Dung beetle

go to Page (26)



In the Assessment Book :

Try to answer :

- Weekly Assessment ①
- [in part ②]

blackish
oval
dung

مائل للسواد
بيضاوي
روت
branches
prominent
network

أفزع
بارز
شبكة
dark green
information

أخضر داكن
معلومات

Questions signed by  are taken from the "Performances and Assessments Book"

First Questions of El-Moasser Book

1 Complete the following sentences:

- ① A is used to observe some living organisms closely.
- ② When looking at a tree trunk using a magnifying glass, move your and back and front until you get a clear image.
- ③ When observing a living organism closely, you can record your observations on a
- ④  When preparing a record card, write the first.
- ⑤ The record card should include sentences that describe, shape,, and location.

2 Put [✓] or [X]:

- ① When observing a living organism closely, we must observe its shape, color, and weight. ()
- ② A record card sketch should use clear, shaded lines. ()
- ③ The dung beetle rolls a ball of dung backward. ()

3 What happens when you move the magnifying glass away from your eye, while observing a living organism closely.

4 How can a living organism be described on a record card?

5 What are the main items that you should write in the record card?

6 Explain how to use a magnifying glass to examine a large rock.

7 In the opposite figures, which of the two girls is using the magnifying glass correctly to observe a leaf closely? Why?

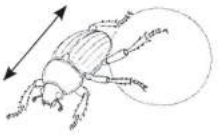


1



2

8 Complete the following record card.

..... (1)			
.....			
			
Size	 (2)	Color	 (3)
Shape	 (4)		
Location	 (5)		

Second Questions of the School Book

A Try:

1 Write [✓] for the correct statement and [X] for the incorrect one about how to use a magnifying glass during observation:

- ① Hold the glass far from your eye and look around. ()
- ② When observing a tree that cannot move, move your own face to focus. ()
- ③ Look at the Sun through the magnifying glass. ()

2 Choose the correct way to sketch for science from below.

- Ⓐ Add shadows to make it look cool.
- Ⓑ Draw the shape exactly as you see it with thin lines.
- Ⓒ If you cannot see a part well, imagine it and draw it.

B Exercise:

1 Write [✓] for the correct statement and [X] for the incorrect one about how you use a magnifying glass during observation:

- ① Hold the glass close to the eye and look at the Sun. ()
- ② To look at a leaf in hand, hold the glass to the eye and move the leaf back and forth. ()
- ③ The object couldn't move, so hold the glass far from the eye and stretch arms out. ()

2 Choose three correct answers:

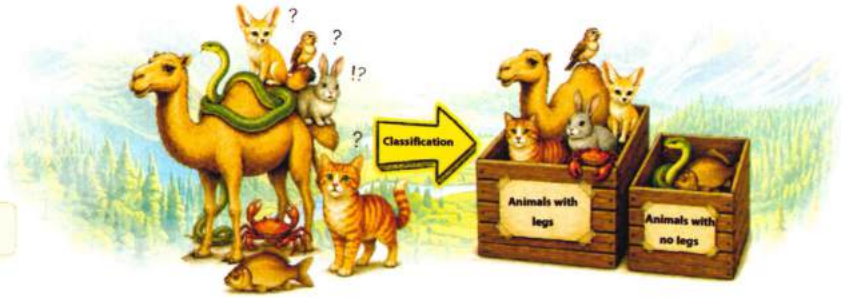
What is important to write on a record card besides the sketch?

- Ⓐ Date.
- Ⓑ Location found.
- Ⓒ What you ate for dinner yesterday.
- Ⓓ Color and shape of the living organism.
- Ⓔ Your favorite colors and shapes.

Colors and Shapes of Living Organisms

- Living organisms can be classified by observing their characteristics, such as:

Color and **Shape** and **Size**



How to Group (Classify) Living Organisms

- To classify living organisms into groups, we must follow the following steps:

What is meant by Classification?

It is the process of collecting living organisms with similar characteristics and placing them in groups.

1

Find

Similar Characteristics

For example

Has legs

Has wings

Similar colors



Different Characteristics

For example

Different shapes

Different sizes



2

Classify

Collect living organisms with similar characteristics and place them in the same group.



Activity

Classifying Some Living Organisms

► Classify the following living organisms into groups by different ways according to:

1 Color

2 Shape

3 Size



Fish



Lizard



Eagle



Butterfly



Grasshopper



Papyrus



Lotus



Lantana



Snake



Spider

1 Color

Brown group



Green group



2 Shape

Group with legs

2 legs



4 legs



6 legs



8 legs



Group with no legs



3 Size

Group >



Group <



lizard
eagle

سحلية butterfly
نسر papyrus

فراشة lotus
نبات البردي lantana

نبات اللوتس
نبات اللانتانا

Questions on Chapter 1

LESSON 3

Answered

Colors and Shapes of Living Organisms

Questions signed by  are taken from the "Performances and Assessments Book"

Questions of El-Moasser Book

1 Complete the following sentences:

- ① Living organisms can be classified by observing characteristics such as and
- ② The spider has legs, while the lizard has legs.
- ③ Living organisms with the same number of legs include and

2 Choose all the correct answers: To classify living organisms into groups, we

- (a) look for similarities.
- (b) place different organisms in the same group.
- (c) look for differences.
- (d) place similar organisms in different groups.

3 What is meant by: Classification:

4 Give a reason: Lotus and papyrus plants can be classified in the same group.

5 Compare: Rabbit and spider (in terms of size compared with a sparrow).

Spider Sparrow

Rabbit Sparrow

6 What characteristic is shared by the butterfly and the eagle?

7 Classify the following living organisms into different groups according to:

- ① Has legs.
- ② Has wings.



①



②



③



④



⑤



⑥

Group with no legs

Group with legs

Group with no wings

Group with wings

8 If you are given the following organisms: (spider – sparrow – butterfly – dog),

Group with no wings

Group with wings

How would you classify them into two groups using the characteristic "has wings"?

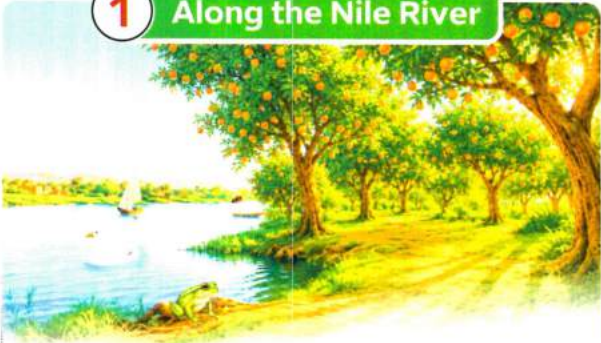
Different Places, Different Living Organisms

The Relation between Habitats and Living Organisms

- ▶ You learned previously that habitat is the place where a living organism lives.
- ▶ The kinds of **living organisms** differ according to their habitats within the **environment** in which they live.
- ▶ Each environment has characteristics that distinguish it and determine the habitats in which living organisms live.

Environment **contains** Habitats **live in each of them** Living organisms

▶ Examples of these environments in Egypt include:

1 Along the Nile River				2 Desert	
					
Amount of water is plenty .				Amount of water is very little .	
Soil is moist .				Soil is dry .	
Amount of sunlight is less .				Amount of sunlight is more .	
Examples of plants					
					
Papyrus	Lotus	Orange	Luffa	Acacia	Colocynth
Examples of animals					
					
Bee	Frog	Goose [water bird]	Jerboa	Fennec Fox	Sand Lizard

distinguish
determine
plenty

يميز moist
يحدد luffa
وقيرة acacia

رطوبة colocynth
نبات اللوف jerboa
نبات السنط

نبات الحنظل
البريوع

Adaptation

- If we move a frog from the Nile River environment to the desert, **What Happens?**
Of course, it **will not** be able to survive, because its body **cannot** withstand the intense heat and scarcity of water.



- For a living organism to live in a place, it must have:

Special characteristics in its body parts.

Special behaviors (ways of living).

What is meant by Adaptation?

It is a change in parts of a living organism's body or in its behavior that helps it survive in a place.

that help it survive in that place. This is known as "Adaptation".

Adaptation of Living Organisms in Egypt's Desert

- In the deserts of Egypt, living organisms have special body shapes and clever ways of living **G.R?** To allow them to survive in this extreme environmental conditions.
Now, we will study:



Adaptation of desert plants



Adaptation of desert animals

To be studied in the next lesson

adaptation
survive
withstand

التكيف intense
يتعايش scarcity
يتحمل behavior

شديدة extreme
قلة/ندرة
سلوك

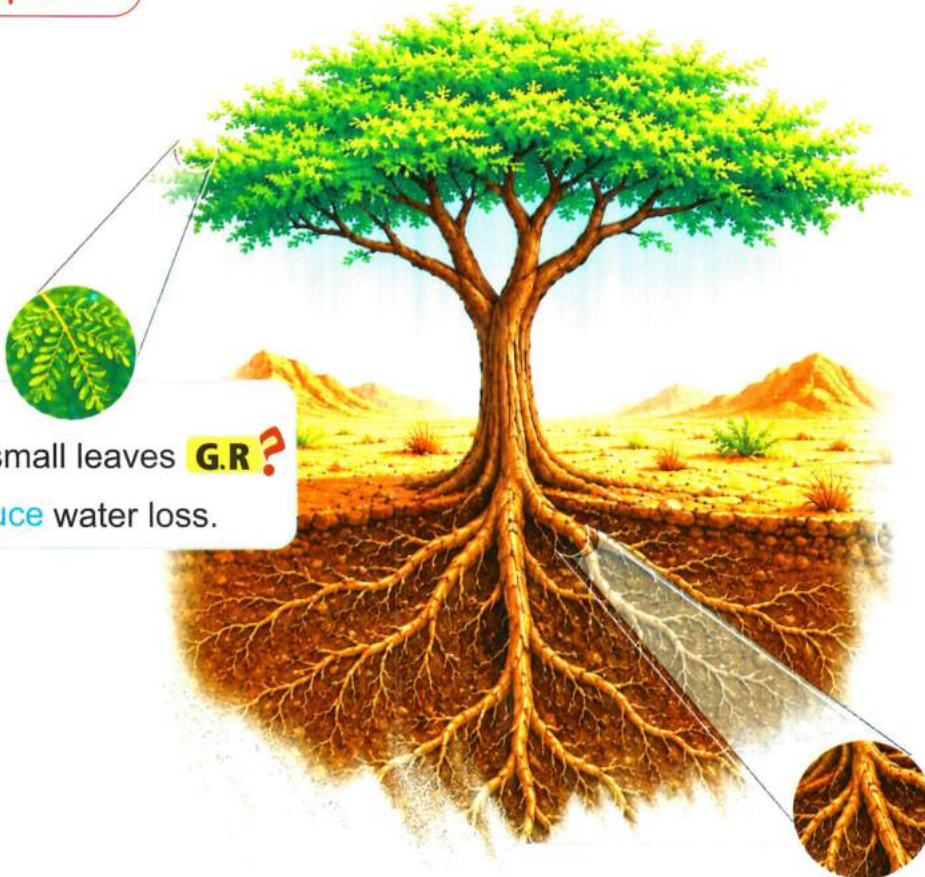
شديد

Adaptation of Desert Plants

- Desert plants have special characteristics that help them retain water.

Examples of these plants include:

Acacia plant



It has small leaves **G.R?**
To reduce water loss.

It has roots that grow deep underground
G.R? To reach the water that is found
deeply underground.

! For illustration only

- Acacia trees use clever ways of living. For example:
If an animal tries to eat their leaves, they begin producing a substance that makes the leaves taste bad.
- Some desert plants, such as cactus, have roots that spread horizontally under the ground to absorb rainwater.

retain
reduce
illustration

يحتفظ بـ taste
يقلل cactus
الإطلاق horizontally

مذاق absorb
الصبار
أفقياً

يمتص

LET'S EXPAND!

► Some living organisms, such as plants and animals, can live in more than one habitat:

First

Plants

1



Lotus

Habitat:
Wetlands
and swamps
along the
Nile River

2



Papyrus

Habitat:
Swamps and
watersides
of the Nile
Delta

3



Egyptian Clover

Habitat:
• Rich lands in:
- The Nile Delta.
- Upper Egypt.

4



Date Palm

Habitat:
• Deserts.
• Oases.
• Places near
the Nile River.

5



Colocynth

Habitat:
Dry deserts
near the
Mediterranean
Sea

6



Acacia

Habitat:
• Dry areas.
• Deserts.
• Around oases.

wetland
swamps
watersides

أرض رطبة
مستنقعات
ضفاف

rich land
oases
Mediterranean Sea

أرض خصبة
الواحات
البحر المتوسط

Egyptian clover
date palm

البرسيم المصري
النخل

1



Hoopoe

Habitat:

- Gardens.
- Farmlands.



2

Cattle Egret

Habitat:

- The Nile River basin.
- Farmlands.
- Green areas.



3



Dung Beetle

Habitat:

- Grasslands.
- Farmlands.
- Deserts.

4

Sand Cat

Habitat:

Deserts east of the Nile River



5



Plain Tiger

Habitat:

- The Nile Delta.
- Valleys.
- City parks.

6

Egyptian Cobra

Habitat:

- Dry semi-desert areas.
- Savannas.
- Woodlands.



In the Assessment Book :

Try to answer :

- Weekly Assessment (2)
- [in part (2)]



hoopoe
cattle egret
basin

الهدهد plain tiger
أبو فردان valleys
ضفاف city parks

فراشة النمر Egyptian cobra
الوديان savannas
حدائق المدن woodlands

الكوبرا
حشائش السافانا
الغابات المفتوحة

Questions on Chapter 1

LESSON 4

Answered

Different Places, Different Living Organisms

Questions signed by  are taken from the "Performances and Assessments Book"

First Questions of El-Moasser Book

1 Complete the following sentences:

- ① The lotus plant grows in while the colocynth plant grows in
- ②  The luffa plant lives in an environment with water that is and soil that is
- ③ Living organisms in Egypt's deserts have special and special of living that help them survive in the environment.

2 Put [✓] or [X]:

- ① The kinds of living organisms that live in any habitat depend on the characteristics of that habitat. ()
- ② The amount of sunlight reaching desert organisms is less than that reaching organisms along the Nile River. ()
- ③ Living organisms do not need adaptation to survive in their habitat. ()

3 Give reasons for the following:

- ① The environment along the Nile River is suitable for growing orange trees.
- ② It is difficult to grow luffa plants in the desert.
- ③ Acacia roots extend deep underground.
- ④  Some desert plants have small leaves.

4 What happens if: A living organism is placed in an environment which is not suitable for its needs.

5 What is meant by: Adaptation:

6 Compare between: The environment along the Nile River and the desert environment (In terms of characteristics).

Desert environment	Environment along the Nile River
.....	

7 Mention the special characteristics of the acacia plant that help it withstand extreme environmental conditions.

8 Give one example of:

- ① A plant grows in desert. (.....)
- ② Birds need an environment includes plenty of water and moist soil. (.....)

Desert Living Organisms

Adaptation of Desert Animals

- Desert environments **such as Egypt's deserts** have extreme conditions such as:

☀ High temperatures. 💧 Water scarcity.

- Desert animals can adapt to the conditions of the desert environment **G.R?**
Because they have special body shapes **and** clever ways of living



First Special Body Shapes

- Desert animals have special body shapes that help them live and survive in their environment, **and examples of these animals:**

1 Fennec Fox

- The soles of its feet are covered with thick fur **G.R?**
To protect its feet from hot sand when walking on it.

- It has big ears **G.R?**
To help it cool its body by allowing excess heat to escape.



Remember

The forest fox's ears are **smaller than** the fennec fox's ears.



Forest Fox



Fennec Fox

soles
escape

باطن
يخرج/يهرب
thick
excess

سميك
الرائدة

2 Jerboa

- It has long back legs **G.R?**

Because it uses them to jump and move quickly, reducing the time its feet touch the hot sand.



Remember

The jerboa is similar to the **kangaroo** because both have long back legs.



Second Ways of Living

- Many desert animals use clever ways to live in the desert, including:

1 Hiding During the Day

- Many desert animals spend the daytime hiding in cool burrows underground **G.R?**

Because the desert sun is intense and the weather is extremely hot during the day.



2 Being Active at Night

- Many desert animals are active at night **G.R?**

Because of lower temperatures at night compared with the intense daytime heat.



3 Getting Water from Food

- Some desert animals rarely drink water **G.R?**

Because they depend on the water in their food (plants or prey) to overcome the lack of water.



4 Drinking a Large Amount of Water

- Some desert animals, such as camels, drink a large amount of water at once, and stores it inside their bodies

G.R? To overcome water scarcity.



From the previous, we conclude that:

- Living organisms adapt to their environment so they can live in it.
- The shape and color of living organisms are adaptations that help them live and survive in their habitat.

kangaroo

spend

daytime

الكينغر

يقضي

فترة النهار

burrow

rarely

depend on

جحر

نادراً

يعتمد على

prey

overcome

weather

فريسة

يتغلب على

الجو/الطقس

Desert Living Organisms

Questions signed by  are taken from the "Performances and Assessments Book"

First Questions of El-Moasser Book

1 Put [✓] or [X]:

- ① Body shape helps the living organism live in its environment. ()
- ② The forest fox's ears are larger than the fennec fox's ears. ()
- ③ Many desert animals hide in burrows during the day to escape from extreme cold. ()

2 Complete the following sentences:

- ①  The feet of the fennec fox have in the soles of its feet.
- ② Many desert animals spend the daytime in cool that are found
- ③ The jerboa has long back legs like the and uses them for and

3 Give reasons for:

- ① Some desert animals depend on food to obtain water.
- ② The soles of the fennec fox's feet are covered with thick fur.
- ③ Desert mice are active at night.

4 What happens if: The fennec fox had small ears.

5 Based on what you have learned, give one example of: A desert animal rarely drinking water and get most needs from their food.

6 Mention the ways of living that help the following living organisms survive in the desert, giving a reason for your answer.



①



②

Second Questions of the School Book

A Try:

- 1 Choose the correct answer: Why are the fennec fox's ears so big?**
- (a) To hear quiet sounds from far away.
 - (b) To let heat inside the body escape to the outside and keep the body cool.
 - (c) To move the big ears like wings and fly in the sky.

- 2 Fill in the blanks with the correct words to explain the reason:**

Why are animals living in the desert not very active during the day?

Because the sunlight is strong, and the weather is very during the day in the desert, animals often rest in cool in the ground or in the shade of rocks.

B Exercise:

- 1 Which of the following is NOT true about the features and life of desert living organisms?**

- (a) Fennec foxes have fur on the soles of their feet to protect them from hot sand.
- (b) Mice in the desert often move around at night to avoid the heat.
- (c) Desert plants spread many big, thin leaves so they do not lose water.

- 2 Fill in the blank about the body feature of the jerboa:**

The jerboa uses its long to jump and move quickly, avoiding the hot ground.



- In the history of Egypt, the scarab (dung beetle) is one of the most famous insects. Ancient Egyptian people treated this bug with great care as a “holy creature that carries the sun.” But, if you look at them through the eyes of science, they have an even more amazing role!



Scarabs Are “Guardians of the Environment”

It helps by:

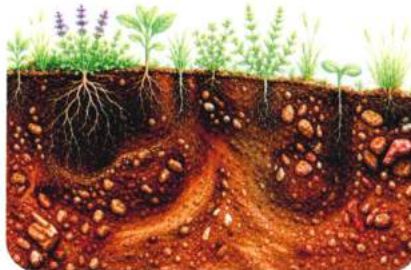
1 Cleaning the Environment

Because it collects animal dung into balls, then rolls, carries, and buries them in the soil.



2 Making the Soil More Fertile

Because when it buries dung in the soil, it adds the nutrients that plants need to grow.



3 Preventing Diseases

Because it helps prevent diseases that could spread from dung left on the ground.



The Amazing Ability of Scarabs

- 1 Recent studies have found that when scarabs roll dung, they are actually looking at the stars. At night, they use the light of the Milky Way as a guide to carry it straight to where they want to go.
- 2 Inside its small body, the scarab has an amazing navigation system that lets it follow the stars as if it carries a natural compass that senses the universe!



⚠ Caution

If you find a scarab in the Egyptian desert, let's watch it without getting in its way. Without this insect, the desert would be a much harder place to live in.





CHAPTER SUMMARY 1

Nature in Our Surroundings

1 Let's Find Living Organisms

- **Living organism:** It feeds, breathes, and grows, like humans, animals and plants, unlike **non-living things** such as rocks, water and air.
- **Habitat:** It is the place where a living organism lives and finds its needs from food, water and shelter.
- **Some habitats and their characteristics:**

Habitat	Properties	Example
1 Near grass or trees	• Food is available. • Easy to hide.	Grasshoppers.
2 Under stones or in the shade of rocks or underground.	• Moist because water is available. • Cool because they are away from sunlight.	Ants.

- **Steps for exploring the habitats of living organisms:**

1 Guess

2 Search

3 Record

2 How to Observe

- **Magnifying glass:** A tool used to help see details of living organisms (such as insects and plants).

How to Use a Magnifying Glass



Can be moved

- Hold the magnifying glass close to your eye.
- Move **the object** back and front to get a clear image.

If the object

Cannot be moved

- Hold the magnifying glass close to your eye.
- Move **your face with the glass** back and front to get a clear image.



How to Write a Record Card

- 1 Write the topic first.
- 2 Write the date, grade, and your name.
- 3 Record what you found using:
 - A sketch.
 - Sentences to describe.
- 4 Write what you learned or write the questions you are thinking about.

Dung Beetle

Name: Grade: Date:

Size: About 2 cm

Color: Black (Blackish).

Shape: Oval-shaped with long back legs.

Location: On the ground near the Nile River.

It rolled a perfectly round ball of dung backward while standing on its head.

3 Colors and Shapes of Living Organisms

- **Classification:** The process of collecting living organisms with similar characteristics and placing them in groups.
- **Living organisms can be classified by observing characteristics such as their colors, shapes, and sizes. This is done by looking for:**

1 Similarities (their similar characteristics).

2 Differences (their different characteristics).

Example

Having legs

Having wings

Similar colors

Example

Different shapes

Different sizes

3 Collecting living organisms that are similar in their characteristics and put them in groups.

4 Different Places, Different Living Organisms

- **Examples of environments in Egypt and their characteristics:**

	environment	characteristics	Examples
1	Along the Nile River	• Water plenty. • Soil is moist. • Amount of sunlight is less.	Plants such as : Lotus – Luffa – Orange – Papyrus. Animals such as : Frog – Goose – Bee.
2	Desert	• Water very little. • Soil is dry. • Amount of sunlight is more.	Plants such as : Colocynth – Acacia. Animals such as : Jerboa – Fennec Fox – Sand Lizard.

- **Adaptation:** A change in a living organism's body parts or behavior that helps it survive in a place.

- **Desert plants have special characteristics such as:**

	Some special characteristics	Importance
 Acacia	Small leaves.	Reducing water loss.
	Roots that grow deep underground.	Reaching water that is found deeply underground.

5 Desert Living Organisms

- Desert animals have **special body shapes** and **clever ways of living** to overcome the extreme conditions:

1 Special Body Shapes

Living Organism	Some special body shapes	Importance
 Fennec Fox	Big ears.	Cooling the body by releasing excess heat.
	Soles covered with thick fur.	Protecting the feet from hot sand when walking on it.
 Jerboa	Long back legs.	Jumping and moving quickly, reducing the time spent touching hot sand.

2 Ways of Living

Way of living	Reason
① Hiding in cool underground burrows during the day.	Because the sunlight is strong and the weather is very hot during the day.
② Being active at night (such as desert mice).	Because the temperature is low at night.
③ Rarely drinking and depending on water found in food.	To overcome water scarcity.
④ Drinking a large amount of water at once and store it inside their bodies (such as camels).	To overcome water scarcity.

Review on Chapter 1

To review this chapter look at the **Assessment Book**
"Part 3 : Final Revision".

In the Assessment Book :

Try to answer :

- Test ① [in part ①]
- Weekly Assessment ③ [in part ②]



Questions signed by  are taken from the "Performances and Assessments Book"

1 Choose the correct answer:

- ① Which of the following is an example of a non-living thing?
(a) Tree. (b) Sparrow. (c) Cactus. (d) Air.
- ② can live in cracks between rocks in deserts.
(a) Sparrows (b) Butterflies (c) Lizards (d) Frogs
- ③ The first step during writing your observation on a record card of a living organism is
(a) writing the topic.
(b) drawing a sketch of the living organism.
(c) writing questions.
(d) writing sentences to describe the living organism.
- ④ During classifying the rabbit and the forest fox, both are similar in their
(a) number of legs. (b) shape. (c) size. (d) length.
- ⑤ lives along the Nile River.
(a) Frog (b) Sand Lizard (c) Jerboa (d) Fennec Fox
- ⑥ grows in the desert environment.
(a) Luffa (b) Acacia (c) Orange (d) Lotus
- ⑦ Which of the following desert animals drinks a large amount of water at once and stores it inside its body?
(a) Desert mouse. (b) Fennec fox. (c) Jerboa. (d) Camel.
- ⑧ Which of the following is true about the fennec fox?
(a) It has long back legs.
(b) It has small ears.
(c) The soles of its feet are covered with thick fur.
(d) It is always active during the day.

2 Choose all the correct answers:

- ① Living organisms live in ways that are adapted to the environment, in hot and dry places
(a) underground. (b) in the shade of rocks.
(c) under stones. (d) near grass.
- ② When observing a living organism closely, you can observe its
(a) shape. (b) color. (c) size. (d) weight.
- ③ The sparrow and The butterfly can be classified in the same group by shape because both are similar in having
(a) legs. (b) the same size. (c) wings. (d) the same color.

- ④ Among examples of living organisms that are found in Egypt's deserts are
- (a) sand lizard. (b) papyrus. (c) date palm. (d) goose.
- ⑤ Which of the following is true about the jerboa?
- (a) It has long back legs. (b) It is always active during the day.
- (c) It hides in cool burrows. (d) It drinks a large amount of water at once.
- ⑥ Which of the following describes the environment along the Nile River?
- (a) Water is available. (b) The soil is moist.
- (c) The soil is dry. (d) Water is scarce.

3 Choose from column (B) what suits it in column (A):

(A)	(B)
1. Small acacia leaves	a. Store water.
2. Acacia roots	b. Reduce water loss.
	c. Reach water under the ground.

1.

2.

4 Put (✓) or (x):

- ① Some living organisms prefer to live near trees because food is available. ()
- ② Grasshoppers can easily hide near grass and trees. ()
- ③ Places under stones and the shade of rocks are moist. ()
- ④ Ants live in dry places under stones. ()
- ⑤ When using a magnifying glass to observe an object that can move, move the object back and front. ()
- ⑥ You can look directly at the sun using a magnifying glass. ()
- ⑦ You should write your questions and observations at the end of the record card. ()
- ⑧ Living organisms can be classified into groups by observing only their colors. ()
- ⑨ The falcon and the butterfly are similar because both have wings. ()
- ⑩ Farmland along the Nile River is the habitat of the jerboa. ()
- ⑪ The dung beetle is an organism that may be found near the Nile River. ()
- ⑫ The jerboa and kangaroo are similar in jumping and moving quickly. ()
- ⑬ Many desert animals are always active during the day. ()
- ⑭ The camel can drink a large amount of water at once to adapt to the desert environment. ()

5 Complete using the following words when preparing a record card:

(sentences to describe – questions – date and name – sketch)

- ① The step after writing the topic title is to write the
- ② What you find about the living organism is recorded using a with thin, clear lines and no shading.
- ③ The characteristics of the living organism include the place, size, shape, and color.

6 Choose the correct answer from those between brackets:

- ① Sparrows can hide (underground – near trees)
- ② When observing an object that can move, the magnifying glass should be placed (close to your eye – far from your eye)
- ③ Many animals in Egypt's deserts hide in burrows during the day to escape from (intense sunlight and heat – extreme cold)
- ④ The help the fennec fox release excess heat. (wide eyes – big ears)

7 Write the scientific term of each of the following:

- ① Everything that feeds, breathes, and grows. (.....)
- ② The place where a living organism lives. (.....)
- ③ A tool used to see details of some living organisms. (.....)
- ④ Grouping living organisms with similar characteristics into groups. (.....)
- ⑤  A change in a living organism's body parts or behavior that helps it survive in a place. (.....)

8 Complete the following sentences:

- ① Living organisms differ from non-living things because living organisms, breathe and
- ② Among the examples of living organisms are and
- ③  Sand, rocks and water in the schoolyard are classified as
- ④ Places under stones or in the shade of rocks are where there is
- ⑤ Lizards can live in hot, dry places such as
- ⑥ The steps for exploring habitats are guess, and
- ⑦ When observing a living organism with a magnifying glass, wear gloves to protect your hands from getting
- ⑧ When using a magnifying glass to observe a tree trunk closely, move your and back and front until you get a clear image.
- ⑨ What you find about a living organism is recorded on a record card using and

- 10 The dung beetle has an shape and a color.
- 11 Living organisms are classified by looking for and among them.
- 12 Papyrus and lotus can be classified in the same group because they are similar in
- 13 According to the shape, the lizard is classified with organisms that have legs, while the grasshopper is classified with organisms that have legs.
- 14 The environment has little water and dry soil.
- 15 The desert is a habitat for many animals, such as, the lizard, and
- 16 The environment is considered as a suitable habitat for orange plant.
- 17 Living organisms in Egypt's deserts have special and that help them adapt to and survive in the environment.
- 18 The jerboa and kangaroo are similar because both have
- 19 Many desert animals are active at night when the temperature is
- 20 The acacia tree has small and that grow deep underground.

9 Give reasons for the following:

- 1 The Sun and water are not considered as living organisms.
.....
- 2  The cat is considered as a living organism.
.....
- 3 The kinds of living organisms differ from one habitat to another.
.....
- 4 Some birds prefer to live near grass or trees.
.....
- 5 Some animals prefer to live in the shade of rocks or under stones.
.....
- 6 The camel and the snake can be classified into two different groups.
.....
- 7 The falcon and the chicken can be classified in the same group.
.....
- 8 The area along the Nile River is a suitable environment for lotus plants.
.....
- 9 Living organisms in Egypt's deserts have special characteristics.
.....
- 10 The camel drinks large amounts of water at once.
.....
- 11 The fennec fox has big ears.
.....
- 12 The soles of the fennec fox's feet are covered with thick fur.
.....
- 13 The jerboa has long back legs.
.....
- 14  Many desert animals are active at night.
.....
- 15  The fennec fox hides in cool underground burrows during the day.
.....

- 16 Some desert animals rarely drink water.
- 17 Desert plants, such as acacia trees, have small leaves.
- 18 Acacia roots grow deep underground.

10 What happens if...?

- 1 A rabbit is moved to live with a camel in the hot desert.
- 2 Living organisms are unable to change their behavior when their environment is changed.
- 3 The leaves of the acacia tree are large.
- 4 The roots of the acacia plant are short and close to the ground surface.
- 5 The surface area of desert plant leaves increased greatly.
- 6 A jerboa is moved to an environment with plenty of water.

11 What is meant by [Define]:

- 1 Habitat:
- 2 Classification:
- 3 Adaptation:

12 Cross out the odd word:

- 1 Rocks – Water – Trees – Air. (.....)
- 2 Frog – Ant – Dog – Cactus. (.....)
- 3 Jerboa – Fennec Fox – Frog – Sand Lizard. (.....)
- 4 Lotus – Luffa – Acacia – Orange. (.....)
- 5 Frog – Bee – Goose – Sand Lizard. (.....)

13 Compare between:

The desert environment **and** the environment along the Nile River (In terms of: Amount of water – Amount of sunlight – Examples of living organisms found there)

Points of comparison	The desert environment	The environment along the Nile River
Amount of water:		
Amount of sunlight:		
Examples of living organisms found there:		

14 What is the importance of each of the following:

- ① The jerboa's back legs:
- ② The deep roots of the acacia plant:
- ③ The small leaves of the acacia plant:

15 Give one example of each of the following:

- ① An animal that lives near grass or trees. (.....)
- ②  An animal that lives along the Nile River. (.....)
- ③ A desert animal with big ears and thick fur on the soles of its feet. (.....)
- ④ A desert plant with small leaves. (.....)
- ⑤ A forest animal with small ears. (.....)
- ⑥ A special body shape that helps the jerboa jump and move quickly. (.....)

16 Variant questions:

- ①  Look at the following figures, then classify them as living organisms or non-living things:



1

(.....)



1

(.....)



3

(.....)



4

(.....)

- ② Classify the following living organisms as plants or animals:

(Acacia tree – Cat – Grasshopper – Ant – Palm tree – Pigeon – Cactus – Luffa)

Animals	Plants
.....	
.....	
.....	

- ③ Ahmed explored a living organism by following these steps. Write the name of the step represented by each picture, then arrange the steps in the correct order :



1

(.....)



2

(.....)



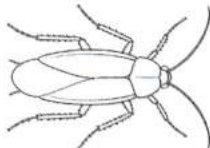
3

(.....)

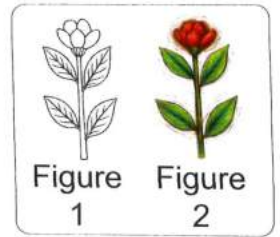
The correct order is

- ④ **How** do living organisms survive in hot, dry places?
- ⑤ **What are** the main characteristics of places under stones?
- ⑥ **How** would you examine a willow leaf with a magnifying glass?
- ⑦ Ahmed and Ayman are completed writing the two record cards after examining an insect in the school laboratory. **Which** of the two cards was recorded correctly using scientific methods **and why?** [Give two reasons].

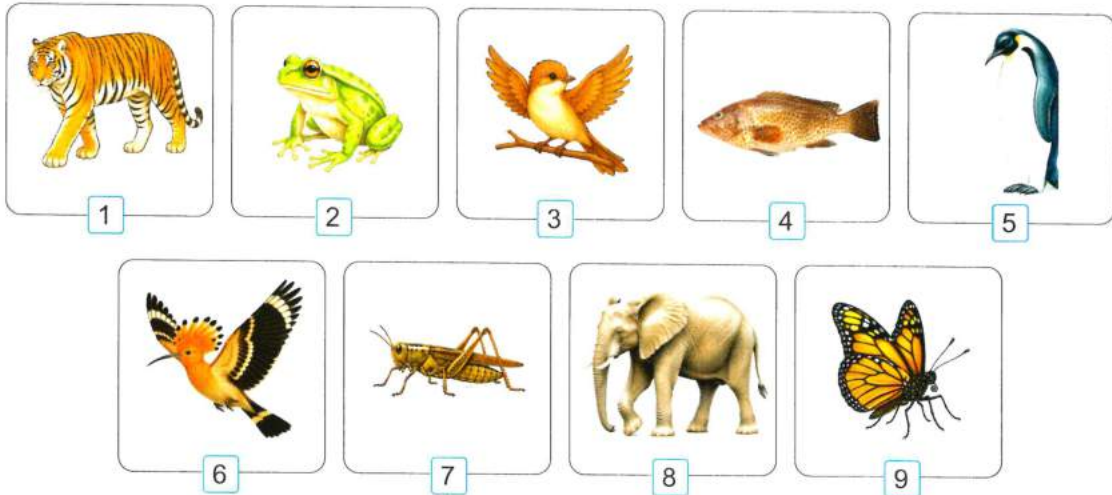
Ayman			
Name: Cockroach	Grade: Four	Date: 21/9/2026	
			
Size	Small	Color	Brown
Shape	Oval body		
Location	Next to the trash bin		
I observed that it is brown, has an oval body, and is small.			
1			

Cockroach			
Name: Ahmed	Grade: Four	Date: 21/9/2026	
			
Size	3.5 cm	Color	Brown
Shape	Oval body		
Location	In the garden		
I observed that it has 3 pairs of legs.			
2			

- 8 Two grade four students observe a plant closely with a magnifying glass. Each student then drew the plant as shown in the two figures. **State which** figure is the correct sketch for a record card. **Give a reason for your answer.**



- 9 **Classify the following living organisms in different ways:**



Record your observations in the following table:

By Color		
By Shape		
By Size		

- 10 **Look at the following living organisms, then state a suitable habitat for each living organism [Along the Nile River – Desert] and the characteristics of this habitat:**

Living Organism	Its Habitat	Characteristics of the Habitat
1 Goose		
2 Lotus		
3 Acacia tree		

11 Look at the opposite two figures, then answer the following questions:

1- **Mention** the name of each organism.

2- **Mention** the environment of each organism.



1



2

12 Complete the table using the suitable statements from below:

- 1- Big ears.
- 2- Spending the daytime in underground burrows.
- 3- Desert animals being active at night.
- 4- Long back legs.

Special body shapes	Clever ways of living
.....	
.....	

Make sure to get

EL-MOASSER

الامتحان

Books

in all subjects

For Grade 4 Primary

